



Department of Computer
Science and Engineering

UTA Smart Kiosk

CSE Senior Design

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Background

The Smart Kiosk is an interactive system designed to modernize how users navigate and interact with campus buildings at UTA. The system enhances accessibility, efficiency, and overall user experience.

System Deployments:

- **Engineering Research South Building A** - A standalone kiosk placed in the lobby.
- **Engineering Research Building** - Two identical kiosk units placed at the main entrances.
- **Remote Version** - Limited Features

Core Features:

- Secure User Login - Card Swipe & Manual Input
- Interactive Map - MappedIn
- Low Supply Requester with Email Notifications
- Conference Room Reservation System
- Slideshow Announcement/Events Panel
- Admin's Dashboard
- Automated Start into Kiosk Mode

Key Requirements

Navigating large campus buildings and accessing essential resources can be inefficient and frustrating for students, faculty, and visitors. UTA is committed to improving accessibility and the overall campus experience. However, several key requirements must be addressed within the ERB and ERSA buildings.

ERSA:

- Establish a structured supply request system
- Implement a conference room identification and reservation system.
- Improve building navigation and wayfinding.

ERB:

- Improve building navigation and wayfinding.
- Provide a centralized information hub for events.



Figure 1. Smart Kiosk in ERSA Lobby

Architectural Design

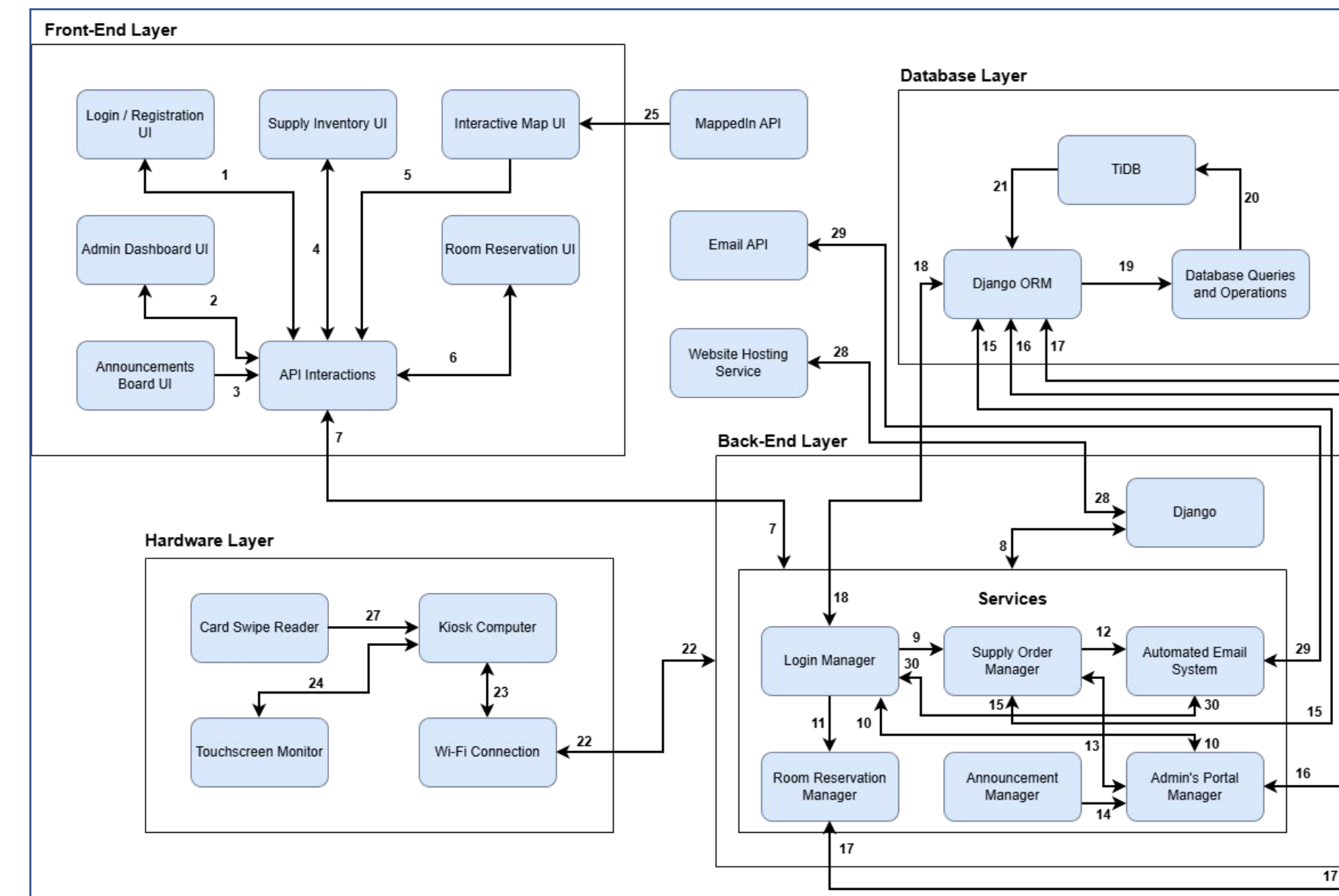


Figure 2. Smart Kiosk Architectural Design Diagram

Table 1. Smart Kiosk Setup Components

Index	Components	Description
1	Frontend UI	React-based touchscreen interface
2	Backend System	Django server handling logic & APIs
3	Database	Cloud-hosted storage for users and data
4	Card Reader	Captures MavID swipe input for user login
5	Display Hardware	Multi-touch touchscreen kiosk interface
6	Network	Enables real-time communication

Implementation Details and Test Plan

Evaluation of the Smart Kiosk is based on a series of usability and performance tests based on real-world user scenarios. Participants included students and faculty members whose feedback was used to improve usability and system reliability.

Test cases:

- User login - card swipe, manual input
- Map - Navigation and Identification
- Conference Room Reservation
- Supply Requests

Performance metrics:

- Task completion time
- System response time
- Error rates during interaction
- User satisfaction and ease of use



Figure 3. ERSA and ERB Kiosk Units

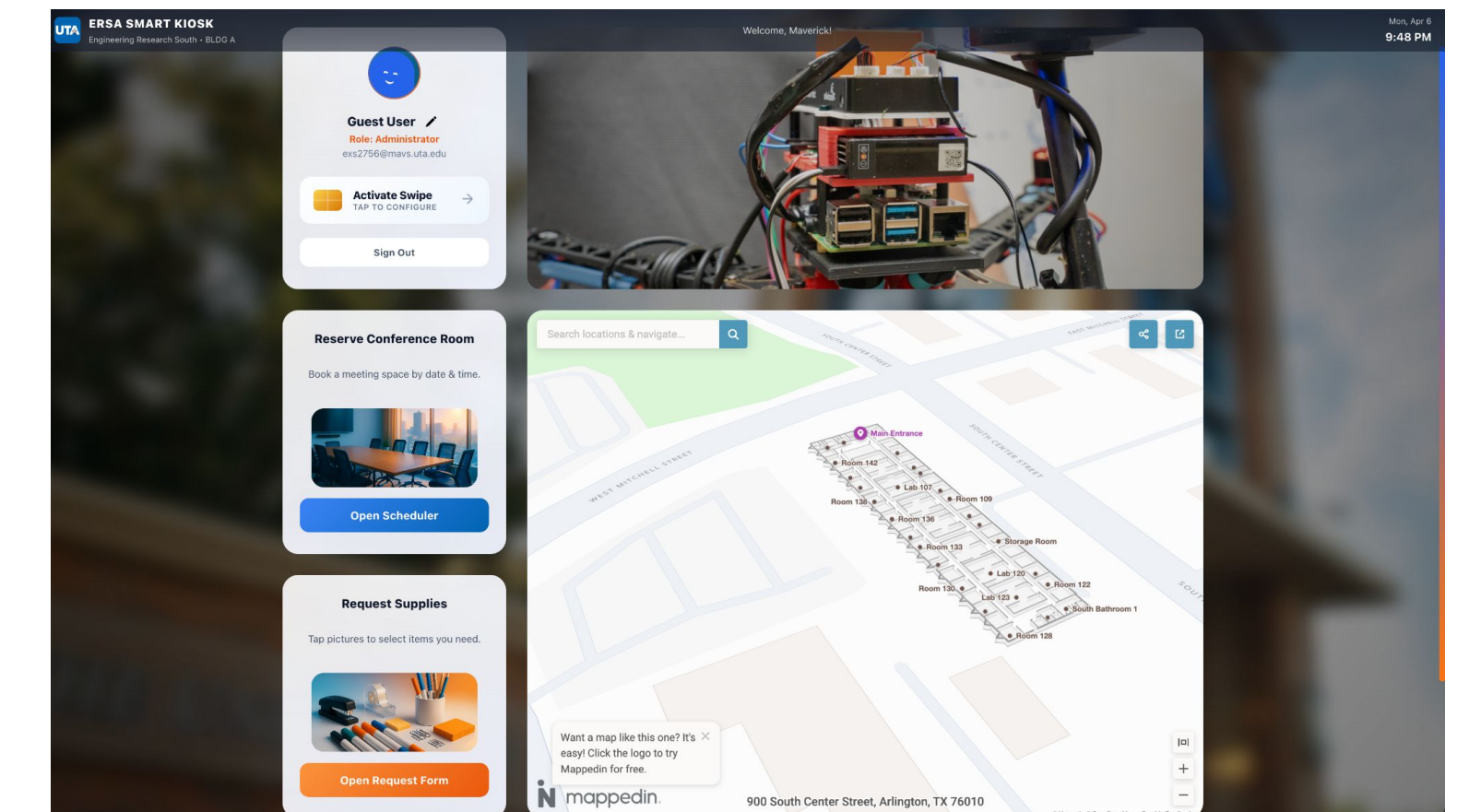


Figure 4. ERSA Kiosk Dashboard

Table 2. Smart Kiosk Evaluation Metrics

Metric Name	Metric Description	Metric Rating
User Rating	Average faculty and student rating based on their experience interacting with the UI	8.2/10
'Page Speed' Accessibility Rating	Performance measure to highlight the accessibility of our web app performed through an automatic AI service.	80%
Swipe Card Login Time	Time it takes for user to log-in to the system using the swipe card system	3.95 seconds
Dashboard Loading Time	Time it takes for the Raspberry Pi to render the ERSA Smart Kiosk Dashboard	3.10 seconds
RPI vs Mini PC performance	Comparison to see how well different kiosk computers run the smart kiosk website. Raspberry Pi vs. KAMRUI Ryzen 7 Mini PC	Mini PC is about 4x more efficient

Conclusions and Future Work

The Smart Kiosk successfully meets the core needs of students, faculty, and visitors by improving navigation, accessibility, and day-to-day convenience within campus buildings. Despite early hardware limitations and delays, our team adapted by prioritizing a strong and reliable software system without sacrificing functionality. The system was built with scalability in mind, allowing it to expand easily to additional campus locations and integrate with additional features. We would like to thank our sponsors, Jade Myers and JD Briggs, as well as Dr. Wang, for their support and guidance throughout this project.

References

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